Analysis: 2023-10-27

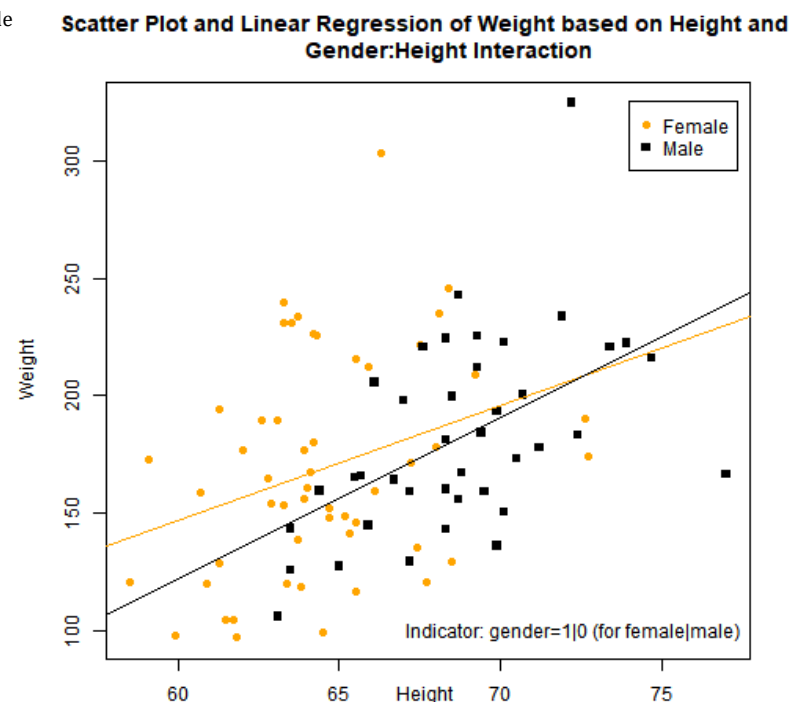
Summary: The purpose of this project is to practice making scatter plots and regression equations with data software.

$$\text{average weight} = \text{intercept} + (\text{slope1})(\text{height}) + (\text{slope2})(\text{indicator: female|male})$$

$$y = b + m_1x_1 + m_2x_2\,, \qquad x_2 = \begin{cases} 1 \text{ if female} \\ 0 \text{ if male} \end{cases}$$

$$y = -214.805 + 5.772x_1 + 11.615x_2$$

Coefficient of Determination = 0.1741

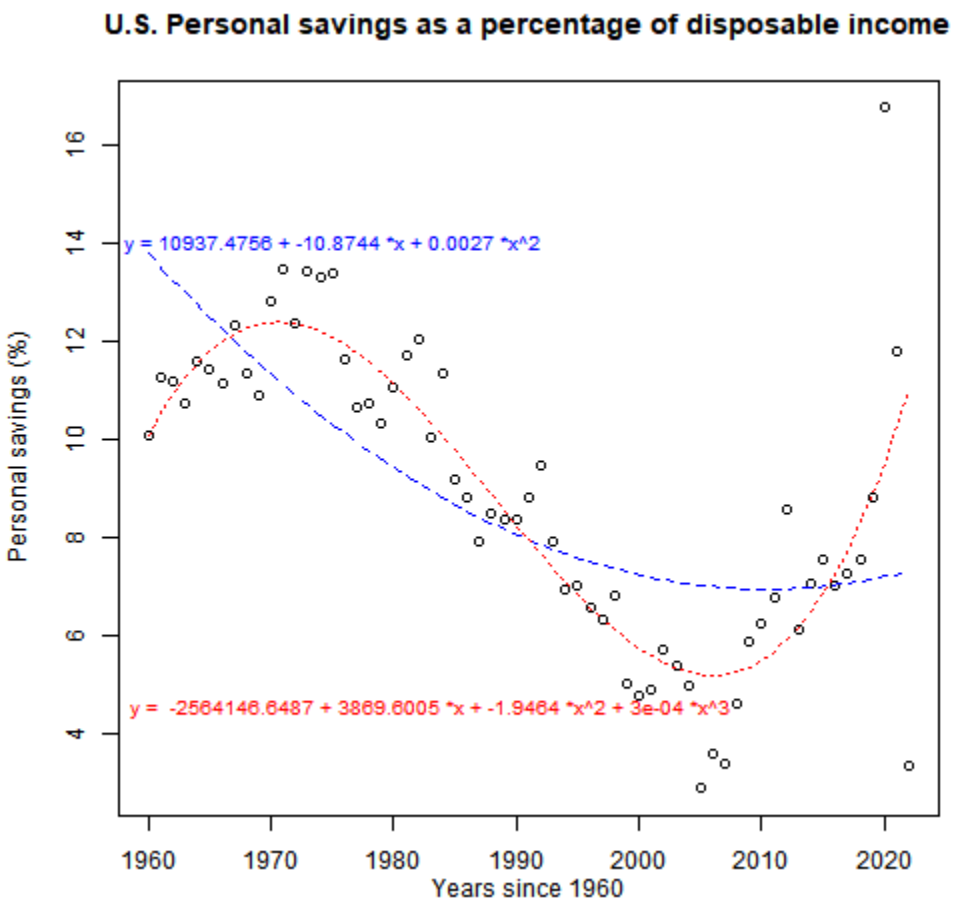


$$\text{average weight} = \text{intercept} + (\text{slope1})(\text{height}) + (\text{slope2})(\text{indicator: female|male}) + (\text{slope3})(\text{indicator: female|male})(\text{height})$$

$$y = b + m_1x_1 + m_2x_2 + m_3x_2x_1\,, \qquad x_2 = \begin{cases} 1 \; if \; female \\ 0 \; if \; male \end{cases}$$

$$y = -289.141 + 6.852x_1 + 145.143x_2 - 1.998x_2x_1$$

Coefficient of Determination = 0.1786



$$\text{average personal savings (\%)1} = \text{intercept1} + (\text{slope1})(\text{year}) + (\text{slope2})(year^2)$$

$$y_1 = b_1 + m_1x_1 + m_2{x_1}^2$$

$$y_1 = 10937.4756286 - 10.8744179x_1 + 0.0027046{x_1}^2$$

$$\text{average personal savings (\%)2} = \text{intercept2} + (\text{slope3})(\text{year}) + (\text{slope4})(year^2) + (\text{slope5})(year^3)$$

$$y_2 = b_2 + m_3x_1 + m_4{x_1}^2 + m_5{x_1}^3$$

$$y_2 = -2564146.6486955 + 3869.6004782x_1 - 1.9464009{x_1}^2 + 0.0003263{x_1}^3$$

$$\text{Coefficient of Determination2} = 0.7166567$$

References

Add Legend to Plot in R. Geeks For Geeks. https://www.geeksforgeeks.org/add-legend-to-plot-in-r/

R plot pch symbols : The different point shapes available in R. Statistical tools for high-throughput data analysis (STHDA). http://www.sthda.com/english/wiki/r-plot-pch-symbols-the-different-point-shapes-available-in-r

Zach. (2020). How to Find Coefficient of Determination (R-Squared) in R. Statology. https://www.statology.org/r-squared-in-r/

```
# Title:   R Script for Mini-Project 02 - Scatterplots ----
# Author:  Kirpatrick Dorsey, MSBA Graduate Student ----
# Date:       2023-10-29

# Note:    This work is best viewed in RStudio, as an R script, using the
#         'outline' view.

# Summary: ....----
#         Provided by Dr. Tristan Whalen, PhD, Professor BUAN 6359.005
#
#         The purpose of this project is to practice making scatter plots
#         and regression equations with data software. You will use R to
#         make a total of three scatterplots, each with a graph of a
#         regression equation.

# --- --- --- --- --- --- --- --- --- --- --- --- --- --- --- --- --- --- --- --- #
# Tech Stack: ----
##           R version 4.3.1 (2023-06-16 ucrt) ----
#           -- "Beagle Scouts"
#           Platform: x86_64-w64-mingw32/x64 (64-bit)
#
##           RStudio 2023.09.0+463 "Desert Sunflower" ----

# --- --- --- --- --- --- --- --- --- --- --- --- --- --- --- --- --- --- --- --- #
# Libraries: ----

## data.table ----
# # data.table _1.14.8
# install.packages("data.table")
library(data.table)

# --- --- --- --- --- --- --- --- --- --- --- --- --- --- --- --- --- --- --- --- #
# Output: See the ./figures directory ----
#        - scatter_wt_basedOn_ht.png
#        - scatter_wt_basedOn_ht_interaction.png
#        - personal_savings_scatter.png
# Create 'figures' directory to store analysis output.
dir.create("figures")

# ____________________________________________________________________________________ ----
# Responses: ----

# Plot 1: from the hanes data,... ----
#   a plot of weight, based on height, and a regression line with an indicator
#   variable for male or female.
#
## Health and Nutrition Examination Survey (hanes)
#  2005 sample, provided by Dr. Whalen.
#  Additional reference: https://www.cdc.gov/nchs/nhanes/index.htm
#
# Read in hanes data. Remove nulls (or na) values. Store gender as an integer.
hanes <- readRDS("hanes.rds")
hanes_clean <- na.omit(hanes)
hanes_clean$genderInt <-as.numeric(hanes_clean$gender=="F")
# hanes_clean$genderInt
# Attach the data set for convenience
attach(hanes_clean)
```

```
# 1. The regression line plus indicator scatter plot.

## a. Make a scatter plot of weight... ----
#   From the hanes data, make a scatter plot of weight (y) based on height (x).
plot(height,weight)


## b. Distinguish the men and the women in the scatter plot... ----
#   ...with two different symbols and colors. Please use entries of the form
#   pch=ifelse(…) and col=ifelse(…) in the plot function to make this
#   distinction.
#
plot(height,weight, pch=ifelse(gender=="F",19,15),
    col=ifelse(gender=="F","orange","black"))
### REF: 'pch' - Statistical tools for high-throughput data analysis (STHDA) ----
# http://www.sthda.com/english/wiki/r-plot-pch-symbols-the-different-point-shapes-available-in-r


## c. Find the regression equation... ----
#   ...of weight based on height plus a gender indicator. Write down (type out)
#   the equation.
#
model1 <- lm(weight~height+genderInt)
# summary(model1)
y_intercept1 <- model1$coefficients[1]
height_coeff <- model1$coefficients[2]
gender_coeff <- model1$coefficients[3]
regression_equation <- paste("Regression Equation: y =",
                        round(y_intercept1,3), "+",
                        round(height_coeff,3), "*height +",
                        round(gender_coeff,3), "*gender")
regression_equation
# "Regression Equation: y = -214.805 + 5.772 *height + 11.615 *gender"


## d. Add a graph of the equation to the scatter plot. ----
#   (it will be two parallel lines). Hint: find the equations of the two lines
#   by substituting 1 or 0 for the indicator. Then use abline twice, same slope,
#   but different intercept.
#
# When gender = 1 (for female)
# y = -214.805 + 5.772 *height + 11.615 *gender
#   = -214.805 + 5.772 *height + 11.615 *1
#   = (-214.805 + 11.615) + 5.772 *height
#   = -203.1897 + 5.772 *height
# or
#   = (y_intercept1 + gender_coeff) + height_coeff*height
#
y_intercept1_female <- y_intercept1 + gender_coeff
slope1_female <- height_coeff # No change to slope
abline(a=y_intercept1_female, b=slope1_female, col="orange")
#
#
# When gender = 0 (for male)
# y = -214.805 + 5.772 *height + 11.615 *gender
#   = -214.805 + 5.772 *height + 11.615 *0
#   = -214.805 + 5.772 *height
```

```
# or
#   = y_intercept1 + height_coeff*height
#
y_intercept1_male <- y_intercept1
slope1_male <- height_coeff # No change to slope
abline(a=y_intercept1_male, b=slope1_male, col="black")


##  e. Paste the plot into your document editor. ----
#    With math typesetting, provide the regression equation and the coefficient
#    of determination below the plot.
#
r_squared <- summary(model1)$r.squared
# REF: Extract R-Squared
# https://www.statology.org/r-squared-in-r/
#
coefficient_of_determination <- paste(" Coefficient of Determination: ",
                                      round(r_squared,4))
coefficient_of_determination
#
# Save Scatter Plot
png(filename = "./figures/scatter_wt_basedOn_ht.png", width = 500,height = 500)
  plot(height,weight,
       pch=ifelse(gender=="F",19,15),
       col=ifelse(gender=="F","orange","black"),
       main = "Scatter Plot and Linear Regression of Weight based on Height",
       xlab="Height \n\n",
       ylab="Weight"
       # sub = paste(regression_equation,"\n", coefficient_of_determination)
  )
  abline(a=y_intercept1_female, b=slope1_female, col="orange")
  abline(a=y_intercept1_male, b=slope1_male, col="black")
  legend(74,325, legend=c("Female","Male"), pch=c(19,15), col=c("orange","black"))
  legend(66,110, box.lty=0,
         legend="Indicator: gender=1|0 (for female|male)",)
  ### REF: Add a Legend to Plot in R ----
  # https://www.geeksforgeeks.org/add-legend-to-plot-in-r/
dev.off()

# --- --- --- --- ---
dev.off(dev.list()["RStudioGD"])
# --- --- --- --- ---


# Plot 2: From the hanes data... ----
#  a plot of weight, based on height, and a regression line with an indicator
#  variable for male or female, plus one interaction term for gender:height.

# 2. The regression line plus indicator plus interaction.

## a. Repeat part 1, but this time add... ----
#  ...the interaction term gender:height to the regression equation.
#  Write down (type out) the equation.
#
# Re-create the base scatter plot, from part1, of weight based on height
plot(height,weight, pch=ifelse(gender=="F",19,15),
     col=ifelse(gender=="F","orange","black"))
#
```

```
# Build the new model to generate the regression equation coefficients,
#  including the coefficient for the interaction term gender:height.
model2 <- lm(weight~height+genderInt+genderInt:height)
# summary(model2)
y_intercept2 <- model2$coefficients[1]
height_coeff2 <- model2$coefficients[2]
gender_coeff2 <- model2$coefficients[3]
genderHeight_coeff <- model2$coefficients[4]
regression_equation2 <- paste("Regression Equation: y =",
                        round(y_intercept2,3), "+",
                        round(height_coeff2,3), "*height +",
                        round(gender_coeff2,3), "*gender +",
                        round(genderHeight_coeff,3), "*gender*height")
regression_equation2
# "Regression Equation:
#       y = -289.141 + 6.852 *height + 145.143 *gender + -1.998 *gender*height"


## b. Add a graph of the equation to the scatter plot ----
#  (it will be two lines, but not parallel). Hint: find the equations of the two
#  lines by substituting 1 or 0 for the indicator and collecting like terms. Then
#  use abline twice, different slopes, different intercepts.
#
# Find equation of the two liines ----
#
# When gender = 1 (for female)
# y = -289.141 + 6.852*height + 145.143*gender + -1.998*gender*height"
#   = -289.141 + 6.852*height + 145.143*(1) + -1.998 *(1)*height
#   = -289.141 + 6.852*height + 145.143 + -1.998*height
#   = -289.141 + 145.143 + 6.852*height + -1.998*height
#   = (-289.141 + 145.143) + (6.852 + -1.998)height
#   = -143.9979 + 4.853928*height
# or
# y = (y_intercept2 + gender_coeff2) + (height_coeff2 + genderHeight_coeff)height
#
y_intercept2_female <- y_intercept2 + gender_coeff2
slope2_female <- height_coeff2 + genderHeight_coeff
#
# When gender = 0 (for male)
# y = -289.141 + 6.852 *height + 145.143 *gender + -1.998 *gender*height
#   = -289.141 + 6.852 *height + 145.143 *(0) + -1.998 *(0)*height
#   = -289.141 + 6.852 *height
# or
#   = y_intercept2 + height_coeff2*height
#
y_intercept2_male <- y_intercept2
slope2_male <- height_coeff2
#
# Add the two lines to the scatter plot ----
abline(a=y_intercept2_female, b=slope2_female)
abline(a=y_intercept2_male, b=slope2_male)


## c. Which line is for which gender? ----
#  Distinguish between them using lty and/or col within the abline command.
abline(a=y_intercept2_female, b=slope2_female, col="orange")
abline(a=y_intercept2_male, b=slope2_male, col="black")
legend(74,325, legend=c("Female","Male"), pch=c(19,15), col=c("orange","black"))
```

```
legend(68,105, box.lty=0,
      legend="Indicator: female=1|0 (for female|male)",)


## d. Paste the plot into your document editor. ----
#  With math typesetting, provide the regression equation and the coefficient
#  of determination below the plot.
#
r_squared2 <- summary(model2)$r.squared
coefficient_of_determination2 <- paste(" Coefficient of Determination: ",
                                round(r_squared2,4))
coefficient_of_determination2
#
# Save Scatter Plot
png(filename = "./figures/scatter_wt_basedOn_ht_interaction.png", width = 500,
   height = 500)
  plot(height,weight,
      pch=ifelse(gender=="F",19,15),
      col=ifelse(gender=="F","orange","black"),
      main = "Scatter Plot and Linear Regression of Weight based on Height and
      Gender:Height Interaction",
      xlab="Height \n\n",
      ylab="Weight"
      #sub = paste(regression_equation2,"\n", coefficient_of_determination2)
  )
  abline(a=y_intercept2_female, b=slope2_female, col="orange")
  abline(a=y_intercept2_male, b=slope2_male, col="black")
  legend(74,325, legend=c("Female","Male"), pch=c(19,15), col=c("orange","black"))
  legend(66,110, box.lty=0,
        legend="Indicator: gender=1|0 (for female|male)",)
dev.off()

# --- --- --- --- ---
detach(hanes_clean)
dev.off(dev.list()["RStudioGD"])
rm(list = ls())
# --- --- --- --- ---


# Plot 3: on your own,... ----
#  find population data for at least ten years for a city, region, state, or
#  country of your choice at statistista.com. Find two different regression
#  polynomials and plot graphs of both.


# 3. Population polynomial

## a. At statista.com, find population data... ----
#  by year for a city, region, state, or country of your choice. Ideally, find
#  one with an interesting trend, where a polynomial rather than a line would be
#  helpful. There must be at least 10 years recorded, up to a recent year
#  (say, 2021).
#
### U.S. Personal savings as a percentage of disposable income ----
# The statistic presents the average personal saving rate in the United States
# from 1960 to 2022.
# The saving rate is defined as the ratio of money saved by individuals or
# families to their disposable income (income after taxes).
```

```
# https://www.statista.com/statistics/246234/personal-savings-rate-in-the-united-states/


## b. Store the data in R. ----
#  The simplest way is to store two vectors (lists of numbers combined with
#  c(…)), one for the years, and another for the population values. The years
#  can be the absolute year numbers or a counting number starting at 0.
#  Alternatively, you could look into storing it as a data.frame. Or, you could
#  type (or download) the data into an Excel file and import the Excel file.
#
# Original Data Import: The information was (1) downloaded as an excel file, and
# then (2) manually copy/pasted/saved to a separate .csv file (since it was a
# small data set).
#
# statista <- fread("statista.csv")
#
# Revised Data Import for no external data import
statista <- data.table(
  year = c(1960,1961,1962,1963,1964,1965,1966,1967,1968,1969,1970,1971,1972,1973,
         1974,1975,1976,1977,1978,1979,1980,1981,1982,1983,1984,1985,1986,1987,
         1988,1989,1990,1991,1992,1993,1994,1995,1996,1997,1998,1999,2000,2001,
         2002,2003,2004,2005,2006,2007,2008,2009,2010,2011,2012,2013,2014,2015,
         2016,2017,2018,2019,2020,2021,2022),
  personal_savings_percentage = c(10.07,11.28,11.2,10.73,11.59,11.44,11.16,
                                  12.33,11.33,10.9,12.8,13.48,12.35,13.45,
                                  13.3,13.41,11.63,10.66,10.73,10.32,11.06,
                                  11.72,12.04,10.05,11.33,9.17,8.83,7.91,
                                  8.48,8.38,8.37,8.8,9.45,7.93,6.91,
                                  6.99,6.56,6.33,6.8,5,4.74,4.9,
                                  5.7,5.38,4.95,2.86,3.56,3.35,4.58,
                                  5.87,6.23,6.78,8.57,6.13,7.05,7.54,
                                  7,7.25,7.55,8.83,16.78,11.78,3.31))
#
attach(statista)
nrow(statista)# 63
min(year)# 1960
max(year)# 2022
statista


## c. Make a plot of the population... ----
#  based on the year.
plot(year,personal_savings_percentage)


## d. You must include an x-axis label, ----
#  something like “Year” or “Years since _____” with the blank filled in with
#  the starting year.
plot(year,personal_savings_percentage,
    xlab="Years since 1960")


## e. You must include a y-axis label. ----
#  This may be “Population” or “Population (thousands)” or “Population
#  (millions)” and so on.
plot(year,personal_savings_percentage,
    xlab="Years since 1960",
    ylab="Personal savings (%)")
```

```
## f. You must include a main title... ---
#  that says what place in the world the data represents.
plot(year,personal_savings_percentage,
    xlab="Years since 1960",
    ylab="Personal savings (%)",
    main = "U.S. Personal savings as a percentage of disposable income")


## g. Find the least-squares polynomial... ----
#  Following the Galileo example in Lesson 06, find the least-squares
#  polynomial of degree two (or higher). Type out the equation.
Year2 <- year^2
model1 <- lm(personal_savings_percentage~year+Year2)
# model1
model1.equation <- paste("y =", round(model1$coefficients[1],4),
                    "+",round(model1$coefficients[2],4),"*x",
                    "+",round(model1$coefficients[3],4),"*x^2")
model1.equation
# "y = 10937.4756286 + -10.8744179 *x + 0.0027046 *x^2"


## h. Add the polynomial to your scatterplot. ----
#  Use curve to add the polynomial to your population scatter plot.
curve(model1$coefficients[1] + model1$coefficients[2]*x
     + model1$coefficients[3]*x^2,
     add=TRUE,
     col="blue",
     lty=2)


## i. Find the least-squares polynomial with one more degree ----
#  (so, if you found a degree 2 polynomial, now find a degree 3 polynomial).
#  Type out the equation.
Year3 <- year^3
model2 <- lm(personal_savings_percentage~year+Year2+Year3)
# model2
model2.equation <- paste("y = ", round(model2$coefficients[1],4),
                    "+",round(model2$coefficients[2],4),"*x",
                    "+",round(model2$coefficients[3],4),"*x^2",
                    "+",round(model2$coefficients[4],4),"*x^3")
model2.equation
# "y =  -2564146.6486955 + 3869.6004782 *x + -1.9464009 *x^2 + 0.0003263 *x^3"


## j. Add this polynomial to the scatter plot... ----
#  using a different col and lty.
curve(model2$coefficients[1]+model2$coefficients[2]*x
     + model2$coefficients[3]*x^2
     + model2$coefficients[4]*x^3,
     add=TRUE,
     col="red",
     lty=3)


## k. Paste the plot into your document editor. ----
#  Paste the plot (with the population dots and graphs of two polynomials)
```

```
#  into your document editor (Word). With math typesetting, provide the
#  equations of the two polynomials below the plot, along with the coefficients
#  of determination for each.
#
# Find the coefficients of determination for each of the previous models.
paste("2nd Degree Coefficient of Determination: ", summary(model1)$r.squared)
paste("3rd Degree Coefficient of Determination: ", summary(model2)$r.squared)
#
# Save Scatter Plot
png(filename = "./figures/personal_savings_scatter.png",
   width = 500,height = 500)
plot(year,personal_savings_percentage,
    xlab="Years since 1960\n",
    ylab="Personal savings (%)",
    main = "U.S. Personal savings as a percentage of disposable income"
    # sub = paste(
    #   model1.equation, "Coefficient of Determination: ",
    #                                summary(model1)$r.squared, "\n",
    #   model2.equation, "Coefficient of Determination: ",
    #                                summary(model2)$r.squared
    # )
    )
curve(model1$coefficients[1]+
       model1$coefficients[2]*x+
         model1$coefficients[3]*x^2,
     add=TRUE,
     col="blue",
     lty=2,
     text(x=1975, y=14, model1.equation,cex=.80,col="blue"))
curve(model2$coefficients[1]+
       model2$coefficients[2]*x+
         model2$coefficients[3]*x^2+
           model2$coefficients[4]*x^3,
     add=TRUE,
     col="red",
     lty=3,
     text(x=1983, y=4.5, model2.equation,cex=.80,col="red"))
dev.off()


# --- --- --- --- ---
detach(statista)
rm(list = ls())
dev.off(dev.list()["RStudioGD"])
# --- --- --- --- ---
```